

CS391 – Entrepreneurship & Technology Commercialization

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Reference Book: *Entrepreneurship: Successfully Launching New Ventures*

6th Edition by Bruce R. Barringer & R. Duane Ireland

Creativity, Innovation, and Idea Generation

Lecture Overview:

Understanding Creativity and Innovation in Entrepreneurship

Creativity Techniques & Tools for Idea Generation

From Ideas to Opportunities - Screening and Refinement

- Learning Objectives
 - Differentiate creativity from innovation and understand their roles in entrepreneurship.
 - Apply structured creativity techniques (Brainstorming, SCAMPER, Design Thinking) to generate tech-based ideas.
 - Screen and refine ideas into feasible opportunities aligned with technology commercialization.
 - Practice evaluating ideas for problem-solution fit and market readiness.

Understanding Creativity and Innovation in Entrepreneurship

1. Creativity

- "The process of generating a novel and useful idea."
- Important because: entrepreneurship begins with new ideas.

2. Innovation

- Definition: "The process of creating something new, which is central to the entrepreneurial process".
- Creativity = Idea generation; Innovation = Commercialization of those ideas.

3. Why Innovation Matters

- Differentiates startups from competitors.
- Creates sustainable competitive advantage.
- Example: Netflix (innovation in distribution model, not just product).

4. Types of Innovation

- Incremental: Minor improvements (e.g., Gmail interface updates).
- Radical/Disruptive: Breakthroughs that redefine industries (e.g., blockchain, Uber).
- Process Innovation: Improving efficiency (e.g., Amazon logistics).
- Business Model Innovation: New ways of capturing value (e.g., Spotify subscription).

5. Sources of Innovation

- Prior experience (e.g., engineers founding hardware startups).
- Cognitive factors -> entrepreneurial alertness.
- Social networks -> mentors, advisors.
- Environmental trends -> Technological, Social, Economic, Political.

6. Innovation & Technology Commercialization

- Research -> Prototype -> Market-ready product.
- Example: Solar panel labs -> Commercial solar power companies.

Class Activities

Case Example: WhatsApp disrupting SMS (low-cost, internet-driven model).

Discussion: Identify one incremental and one disruptive innovation being used daily.

Creativity Techniques & Tools for Idea Generation

1. Idea Generation Funnel

- Divergent Thinking -> broad idea pool.
- Convergent Thinking -> filtering & selecting promising ones.

2. Creativity Techniques

1. Brainstorming

- Rules: No criticism, encourage wild ideas, build on others.
- Goal: Quantity first, quality later.

2. SCAMPER

- Substitute, Combine, Adapt, Modify, Put to another use, Eliminate, Reverse.
- Example: Food delivery app -> Add subscription -> new revenue model.

2. Creativity Techniques

3. Mind Mapping

- Visual connections -> Example: Smart Home -> security, health, energy.

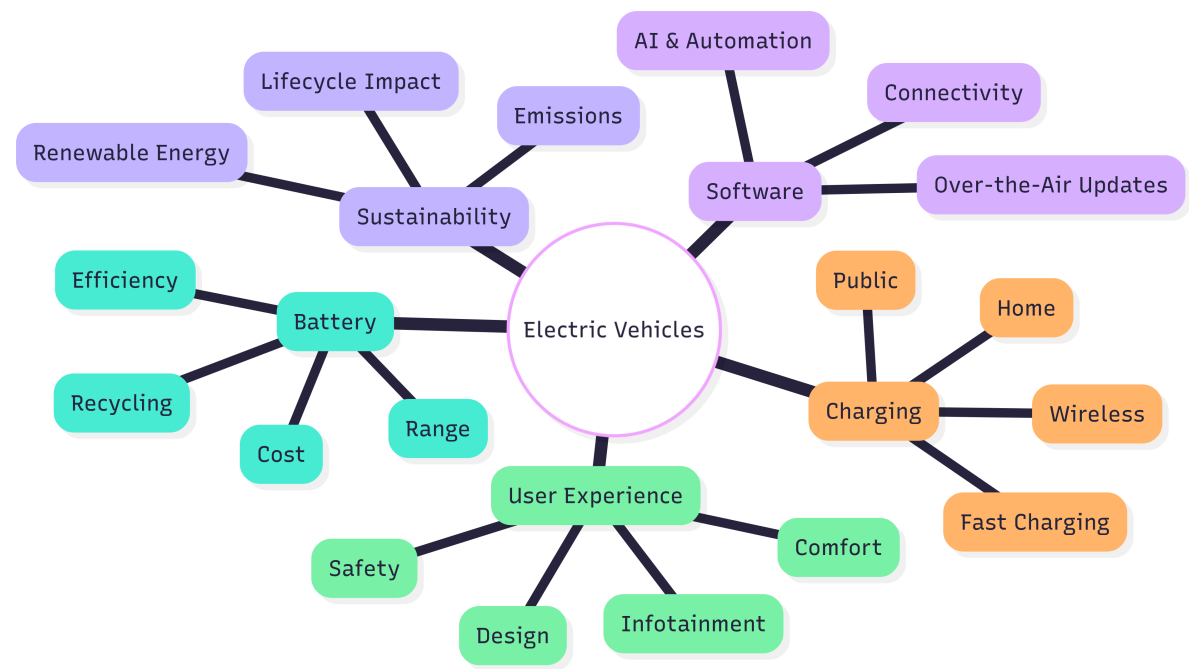
4. Design Thinking (Modern addition)

- Empathize -> Define -> Ideate -> Prototype -> Test.
- Strong fit for technology commercialization.

SCAMPER Example - Focus Point: Smartphone

SCAMPER Step	Application	Example Startup Idea
Substitute	Replace physical SIM with eSIM	Eco-friendly phone with no SIM slot, fully digital connectivity
Combine	Integrate phone with smartwatch (shared data ecosystem)	Smart watch and app for integration with any phone
Adapt	Use gaming phone cooling fans in normal phones.	Compact cooling gadgets that dont make the phone bulky but provide cooling value
Modify	Foldable screens	A second screen gadget attachable to any phone
Put to Another Use	Use old smartphones as IoT home security cameras	Custom rig designing kit
Eliminate	Remove charging port (wireless only)	Fully wireless charging phone - no cables ever needed
Reverse/Rearrange	Instead of phone as personal device -> communal family hub (like Amazon Echo Show)	A new phone OS or something similar

Mind Mapping Example - Future of Electric Vehicles (EVs)



3. Individual vs. Group Creativity

- Individual = originality, depth.
- Group = diversity, cross-fertilization.
- Best practice: start alone, then merge into group ideation.

How These Apply to Tech Commercialization

- Use SCAMPER to repurpose university research into products.
- Apply Design Thinking to refine prototypes (IoT device -> smart health monitoring).

Team Activities

- Brainstorming Drill: 5 minutes to list 30 business ideas for “improving campus life.”
- SCAMPER Exercise: Consider one of the following focus points and perform SCAMPER on that.
 - LLMs
 - Travel and Transport
 - Foods and Restaurants
 - Lifestyle
 - Entertainment
 - Survival
 - Health
 - Finance

Submit your report on MS Teams

From Ideas to Opportunities - Screening and Refinement

1. Problem-Solution Fit

- Is the problem real and painful?
- Does the solution significantly improve over existing alternatives?
- Example: Zoom solved remote meetings better than Skype.

2. Market Pull vs. Technology Push

- Market Pull -> based on demand (e.g., green energy).
- Technology Push -> lab-driven (e.g., graphene applications).
- Best ventures align both.

3. Opportunity Screening Tools

- Product/Service Desirability.
- Industry Attractiveness.
- Market Size & Growth Potential.
- Founder-Team Fit.
- Intellectual Property (protectability).

4. Market Gap Analysis

- Identify competitors.
- Find underserved niches.
- Example: Ride-hailing -> Uber (urban mobility), Careem (regional focus).

5. Business Model Canvas

- Use BMC to quickly test feasibility of new ideas before deep planning.

6. The Technology Lens

Where do tech ventures, specifically those built on data, find their ideas?

1. Automating Tedious Tasks: Look for processes that are manual, repetitive, and rule-based. Data science can often automate them.
 - Example: UiPath (Robotic Process Automation), Zapier.
2. Augmenting Human Decision-Making: Find areas where experts make complex decisions with incomplete information. Build tools that provide predictive insights.
 - Example: Grammarly (augments writing), Tableau (augments data analysis), medical imaging AI (augments radiologists).

6. The Technology Lens

Where do tech ventures, specifically those built on data, find their ideas?

3. Creating New Data Products: Use unique or proprietary data to create a sellable asset or service that didn't exist before.
 - Example: Climate Corporation (selling hyperlocal weather and field-level yield predictions to farmers), Glassdoor (selling insights from employee data to recruiters and companies).
4. Personalizing Experiences at Scale: Use individual user data to tailor products, services, and content.
 - Example: Netflix recommendations, Spotify's Discover Weekly, Stitch Fix's personalized styling.

7. The Nightmare Pitch

- Instead of pitching why an idea is great, you pitch why it's a terrible idea. This forces critical thinking and helps identify fatal flaws early.
- Highlight that the flaws they identified (e.g., "no one would pay for this," "it's too easy to copy") are exactly the kinds of risks

Reading

- Chapter 3: Identifying and Evaluating Entrepreneurial Opportunities -> creativity as a driver of opportunity.
- Chapter 5: Industry and Competitor Analysis -> innovation as a response to change.
- Chapter 13: Preparing for and Evaluating the Challenges of Growth (innovation linked to scaling).
- Chapter 3: Opportunity Recognition (techniques for generating ideas).
- Appendices & Cases -> Examples of brainstorming outcomes in startups.
- Chapter 3: Feasibility Analysis & Opportunity Recognition.
- Chapter 4: Feasibility Analysis.

Acknowledgement

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